Password Attacks: Cracking Protected Files

Objective

The objective of this lab is to crack the password of a protected SSH key file, demonstrating how encrypted files can be vulnerable to password attacks.

Steps

1. Displayed an Encrypted id_rsa File

The lab began by showing an encrypted id_rsa file, which is an SSH private key protected by a password.

```
kira@nix01:~/.ssh$ cat id_rsa
-----BEGIN RSA PRIVATE KEY-----
Proc-Type: 4,ENCRYPTED
DEK-Info: AES-128-CBC,F1C2E21F3CF7BDF460FB56C7D16911F2

sqXnpt6fN4Ugi545CGyPWgfkaQhkDt5lKU6azI4amQ9mifdUkKzdR46EdrU3Pglh
xz3sC+Xdm7qkrtLEQ7rpk8w7zANcsxvQznGspuUv+c1hSvJdVgZAqTG84KFmUpXM
2YW9QdMynHy9PVJP66tKBfuzJ9YZBJISocUjpwtEqrWDnOSBinLsYj6Z+J+yIvLv
1IaFtgGJbqWovJ+5FP9L8Js/AdWRs0fnGzvHYo8h9rJXgf6Qdf7okSsN+pRTPpBe
Dc/IUMwcTUdzyRIMdoz5CsZJGe6jaKYoOt66dU92XjKzHG4yhIeord7+wa1W6MRd
aa2fvvi1SMtaaLFCb6nd8raqoaELPzwcrYQ0cLCwlwggH+GTW/cfVAABnE0kXCgH
sp2uc/KOV6PPdZ6tBFEg/xVehP9H19jd/LfVQ3/tjvKhTNuiF0JcwObiy3d4tpX1
```

2. Converted id_rsa to ssh.hash

The encrypted id_rsa file was converted into a ssh.hash format, making it suitable for password cracking.

```
┌─[eu-academy-6]─[10.10.15.128]─[htb-ac-1246556@htb-f2louus8qw]─[~/Pentest]
└──╼ [★]$ ssh2john.py id_rsa > ssh.hash
┌─[eu-academy-6]─[10.10.15.128]─[htb-ac-1246556@htb-f2louus8qw]─[~/Pentest]
└──╼ [★]$ ls
id_rsa  ssh.hash
```

3. Cracked the Password for the SSH Key

Using a password-cracking tool, the password for the id_rsa file was successfully cracked, allowing access to the SSH key.

```
┌─[eu-academy-6]─[10.10.15.128]─[htb-ac-1246556@htb-f2louus8qw]─[~/Pentest]
└──╼ [★]$ john --wordlist=/usr/share/wordlists/rockyou.txt ssh.hash
Using default input encoding: UTF-8
Loaded 1 password hash (SSH, SSH private key [RSA/DSA/EC/OPENSSH 32/64])
Cost 1 (KDF/cipher [0=MD5/AES 1=MD5/3DES 2=Bcrypt/AES]) is 0 for all loaded hashes
Cost 2 (iteration count) is 1 for all loaded hashes
Will run 4 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
L0veme           (id_rsa)
1g 0:00:00:00 DONE (2024-08-18 05:59) 1.369g/s 2907Kp/s 2907Kc/s 2907KC/s L112893..L0VE13
Use the "--show" option to display all of the cracked passwords reliably
Session completed.
```

Conclusion

This lab demonstrated the process of cracking passwords for protected files, specifically focusing on an encrypted SSH key. The successful decryption of the key highlighted the vulnerabilities associated with weak passwords and emphasized the importance of strong encryption practices.